



A Socially Interactive Robot: MakiMate

Winter 2026

Group #9

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Abstract

Makimate is a socially interactive robot developed to support natural human-robot interaction within a variety of situations. The broader vision of this project is to make social robotics practical and accessible to all. Based on the open-source Maki material, this project aims to improve the robot's capabilities by enhancing speaker identification, developing the robotic arm and torso, and adding new modular behaviors. To achieve this, the electrical, mechanical, and software subsystems seamlessly work together to sense the user, process information, and physically respond. The complete robot assembly integrates a 3D-printed structure, Raspberry Pi 5, camera, microphone, speaker, Dynamixel servo motors, and a ROS2 framework. The finalized system was able to improve user tracking by using multi-modal identity fusion, giving it the ability to reliably focus on the active speaker in noisy or crowded environments. Additionally, a fully engineered robotic arm and torso redesign were established, equipping the robot with tablet integration and expressive gestures. Lastly, improvements to the large language model increased the robot's responsiveness and prepared Maki to serve as a building greeter.



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1. Introduction

MakiMate is a socially interactive robot designed to enable more natural interactions between humans and robots across a variety of applications. The overarching goal of this project is to make social robotics both practical and widely usable. Maki is an open-source platform which allows the team to focus on enhancing the robots performance through improved speaker identification through facial and vocal recognition, the development of a robotic arm and torso, and the addition of an entirely open-source and local conversational pipeline powered by a large language model (LLM).

To accomplish this, the electrical, mechanical, and software subsystems operate together to detect user input, process information, and generate audible and physical feedback. The fully assembled system combines a 3D-printed body with the Raspberry Pi 5, camera, microphone, speaker, Dynamixel servo motors, and ROS2 based architecture.

The final system demonstrated improved user tracking through combined voice and face identification, allowing the robot to consistently identify and follow the active speaker even in noisy or crowded environments. In addition, a newly designed arm and torso system was implemented, enabling tablet integration and more expressive movements. Enhancements to the conversational interface improved response times, expanded the information pool through integrated databases, and made MakiMate less robotic to interact with. All together, these improvements positioned the robot as more approachable and pleasant to interact with, which was the primary goal.



2. Design Specifications

2.1 Mechanical Specifications

Design arms and torso for MakiMate. The initial goal of the arms was to be capable of grabbing and lifting small objects up to two pounds, however, the focus shifted to become more expressive and communicative rather than task oriented. The new goal of the arms was to improve MakiMate's engagement with users by performing gestures when speaking. It was determined that four degrees of freedom would allow for the arms to perform necessary gestures: shoulder pitch and yaw, elbow flexion, and wrist rotation. The design should be able to carry out these motions with ranges of 180° and 75° for the shoulder yaw and pitch, and 75° and 180° for the elbow flexion and wrist rotation. In order to perform these movements, the servo motors driving the motion must have stall torques of at least five times higher than 82Nm, 80Nm, 0.27Nm and .02Nm for the shoulder yaw and pitch, elbow, and wrist, respectively. To go along with the functional requirements, MakiMate must also be visually appealing to users, so the arm must be smooth, round, and proportional [5], and maintain safety standards of limiting sharp edges and pinch points.

2.2 Software Specifications

The original design proposal had described a pipeline built with Pipecat AI, Vosk STT, and Piper TTS, with the goal of integrating speech recognition and LLM-provided dialogue into the existing ROS2 framework. The following changes were made during development.



1. Pipecat AI eliminated since there was no need to put the final conversational pipeline into Piepcat to optimize latency. The combination of the software choices and offloading computing to the server created a fast enough response time to negate the need for Pipecat.
2. Vosk was no longer being used for STT, only for speaker embeddings. The mentors suggested switching from Vosk to another model for STT, as the last group used it, and the results were not acceptable. Through testing, it was determined that Faster-Whisper provided much better models when it came to word error rate. Vosk was retained within the pipeline for speaker embeddings, as Faster-Whisper does not provide this feature.
3. Remote server offload added for ASR. The original proposal did not suggest that ASR would have the potential to be offloaded to a remote server. While the ASR is able to be run on the Pi, larger models would exceed its limitations. In order to provide improved speech recognition, the mentors requested the capability of offloading ASR to a remote server.

2.2.1 Overall System

The software system shall run entirely on the Raspberry Pi 5 without requiring a GPU or cloud services. All processing nodes shall be implemented as ROS2 components that communicate over topics. The full software stack shall start within 10 seconds of powering on the system. The system shall support offloading the ASR and LLM workload to a remote server via HTTP.



2.2.2 Speech Recognition (ASR)

End to end transcription latency shall not exceed 1.5s. The ASR node shall support all Faster-Whisper models sizes, which can be selected via the dashboard. The system shall discard any known Whisper hallucinations and noise-based utterances before publishing. The microphone shall be muted during TTS playback to prevent the robot's own speech from being transcribed.

2.2.3 Vision and Face Tracking

The face detection pipeline shall process camera frames at a minimum of 10 fps on the Pi 5. The system shall detect and track up to 3 simultaneous faces per frame. Tracked face IDs shall persist across frames with no random identity flip. The face recognition system shall correctly identify a registered individual in at least 85% of frames under normal indoor lighting at a distance of 2-4 ft.

2.2.4 Speaker Recognition and Identity Fusion

The voice speaker recognition system shall identify a registered speaker with a similarity threshold of 0.30. Voice speaker profiles shall be captured during a calibration session of no more than 15 seconds per speaker. The identity fusion system shall combine voice and face signals and provide the speaker's name into the LLM prompt when fused confidence meets or exceeds 0.50.



2.2.5 Command Routing and Behavior

The system shall transition from the asleep state to the awake state within 500 ms of the wake phrase being detected. The router shall not forward any utterance to the LLM while a calibration workflow is active.

2.2.6 Dashboard

The web dashboard shall be accessible from any device on the local network at <http://maki.local:8080> without requiring any installation. Robot stdout and ROS2 log output shall be streamed to the browser in real time. All ROS2 parameters must be configurable via the dashboard. The robot shall be able to be operated and developed from the dashboard.

2.2.7 Servo Control

Exponential moving average smoothing shall be applied to all joints to eliminate jittery movement. Hardware and software angle limits shall be enforced to prevent destroying the servos. The smoothing factor shall be tunable via the dashboard.

3. Design Overview

The focus of this project is to build upon a MakeMate robot that was simply put together in the previous senior design class. There are 3 main deliverables of our Senior design project. The first is to create a hardware and/or software system that allows the Maki Robot to distinguish who is speaking within a crowded, noisy, and public environment. The second is to design and



CAD design arms and a torso for the MakiMate. Third is to create a language model where MakiMate can listen to a person, process this and provide verbal feedback.

The mechanical design features an arm depicted in Figure K that is approximately 10 inches in length and bubbly in shape. The arm carries four degrees of freedom, shoulder pitch and yaw, elbow flexion, and wrist rotation. The shoulder motions are carried out by motors housed in the torso and shoulder joint themselves, as shown in Figure 9. The shoulder then connects to the elbow (Figure 8) which features a single motor, as does the wrist (Figure 8), which is connected to the elbow by the forearm. Each of these sections are fastened together to constrain motion to just the desired revolute joints driven by the motors. Wires inside the arm supply the motors with power.

4 . Mechanical Subsystem

Building upon the existing MakiMate model, the goal of this project was to enhance the mechanical subsystem's abilities to perform its intended function of being a sociable robot. This was accomplished by a complete torso redesign that is capable of supporting functional arms, and the integration of a 11-inch tablet. To accomplish this the torso had also been expanded from the original model from 7.2 inches tall, 5.4 inches wide, and a 4 inch depth to 13 inches tall, 8 inches wide and a depth of 5 inches. These dimensions were determined from previously set ratios based on the current head model. It had been determined that having a torso height close to 1.5x the height of the head and a shoulder width 1.5x the width of the head would be considered aesthetically cute and friendly based on the body and head proportions.

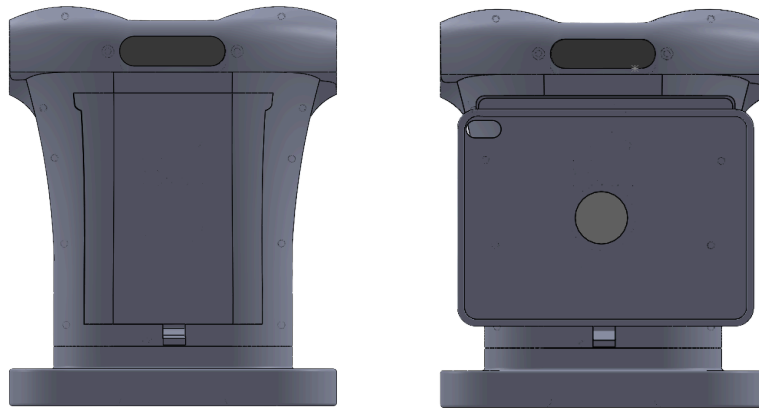


Figure 1: Models of Torso With and Without Tablet

The dimensions of the redesign ended up being close to 1.5x the height of the head (if the base is neglected) and 1.5x the width of the head at the shoulders. This model had also been created with a human-like aesthetic in mind where there is a prominent shoulder and the body tapers down towards the base of the robot (shown above). The additional room created from this redesign allows for the arms to be supported securely, and the ability to add more components without concern for space. The torso is made up of three main shelled pieces that are interlocked with tongue and groove edges and fastened using M5 screws with brass heat set inserts. An internal component tower had also been created to keep electrical components secure and keeping wiring manageable.

The addition of the tablet introduced a unique design challenge. Few social robots currently integrate tablets directly into their structure, making it difficult to determine the most effective implementation method. The initial concept involved embedding the tablet holder directly into the torso shell. However, concerns regarding potential theft in public spaces and

users who choose to not utilize the tablet feature arose. In those cases, leaving an empty holder on the front of the robot would negatively affect the overall appearance.

As a result, the final design incorporated a removable tablet holder. This approach allows consumers who choose not to purchase or use a tablet to install a fitted insert that fills the opening, preserving the robot's clean aesthetic while maintaining design flexibility.

Once the tablet was made removeable, the next challenge was determining how to mount it securely while still allowing a tool-less removal. An initial concept involved a front sliding mechanism, but this approach was not optimal. Although this allowed for easy removal, it made it difficult to conceal the feature when the tablet was not in use.



Figure 2: Tablet Insert and Torso Cover

This led us to the final design (shown above), which utilized a horizontal tablet case with a spring clip at the bottom and an extended feature at the top to secure it within the torso shell. This method was ultimately adopted, and the same spring clip concept was later applied to a torso cover that maintained the overall aesthetic design.

Another task for this project was the addition of a depth camera to provide 3D perception, allowing the robot to understand, navigate, and interact with the physical world and humans in real time, improving its ability to be a sociable robot. The camera that was chosen for this task was the CS30 RGBD ToF Depth Camera by Synexens (shown below).



Figure 3: Synexens CS30 Depth Camera

This camera's mount was integrated into the front shell of the new robot and sits above the new tablet mount. The addition of this camera will take over the task of the 2D camera that is currently located in the head of the Maki. Future iterations of the robot will need to account for this and remove the camera from the head.

After determining the desired functional and aesthetic goals of the arm design using Table 1 to identify the required DOF to perform the gestures, concept sketches (Figure 4) were created to create a foundation for the design. These sketches allowed for visualization of the arm next to components with established dimensions (such as the head), as well as an idea of how servo motors will provide the desired degrees of freedom (DOF). CAD models were also created in SolidWorks to provide shells reflecting the desired size and shape of the arm. These shells (shown in Figure 5) provided a base model from which future iterations could be based off.

Table 1: Degree of Freedom Analysis Chart				
Gesture	Shoulder Pitch (up/down)	Shoulder Yaw (in/out)	Elbow Flexion	Wrist Rotation
Wave	Y	Y	Optional	Optional
Point	Y	Y	Y	N
Beat gesture	Y	N	Optional	Optional
Open arms	Y	Y	N	Y
Closed posture	Y	Y	N	N
Basic raise	Y	N	N	N

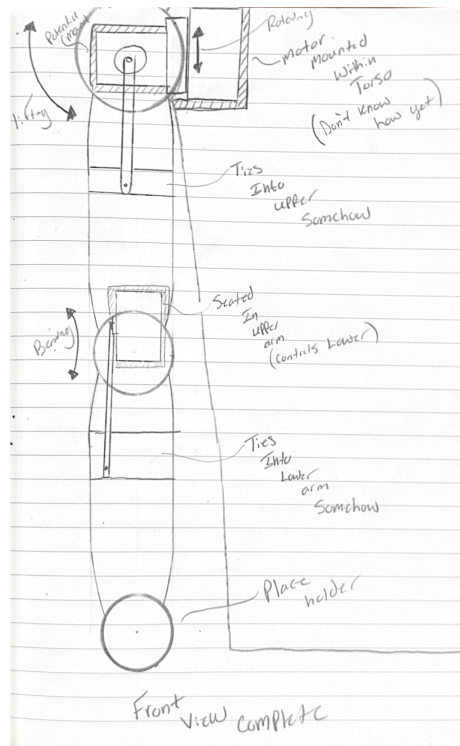


Figure 4: Initial Concept Sketch

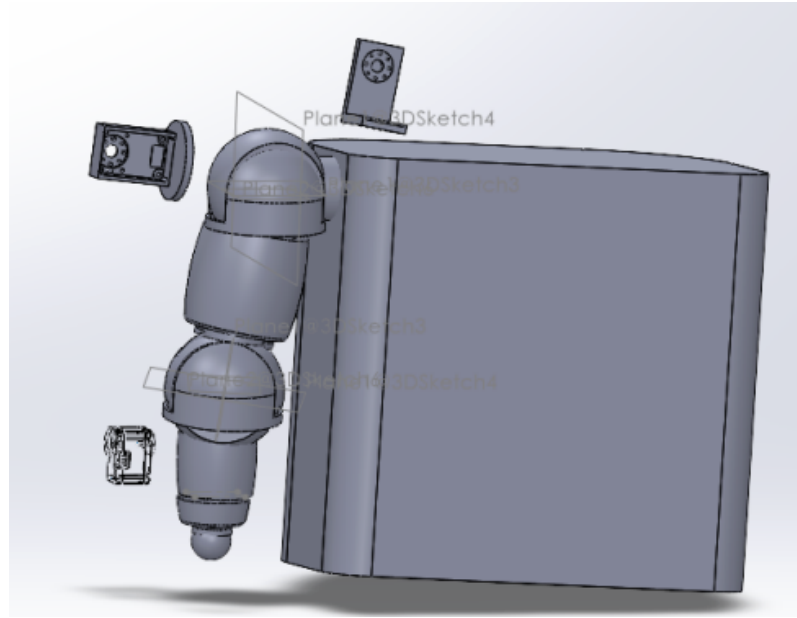


Figure 5: Initial Arm Shells

To decide the motors that would be used to drive each DOF, initial torque requirement estimations were carried out for each servo featured in the arm design using Equations 1 and 2. When considering these torque requirements alongside very rough estimations of factors such as size, weight, and cost (Table 2), the initial servo motors selected were Dynamixel XL430s shown in Figure 5 (for the shoulders) and Dynamixel XL330s shown in Figure 7 (elbow and wrist) as they would remain in the Dynamixel family and allow for daisy chains to ensure serial communication while supplying the smaller torque. The arm design process proceeded with these servos in mind.

$$\tau_{\text{static}} = \sum(m_i g r_i) \quad (1)$$

$$\tau_{\text{design}} = \tau_{\text{static}} * 2 \quad (2)$$

Table 2: Servo Decision Pugh Matrix					
Criteria	Weight	FT5320M (Baseline)	FT5310M	DYNAMIXEL XL430-W250- T	DS3218
Torque/ FOS	5	0	-5	5	5
Size	3	0	3	0	-3
Weight	2	0	0	0	-2
Cost	2	0	2	-2	0
Weighted Score		0	0	+3	0



Figure 6: DYNAMIXEL XL430-W250-T [6]



Figure 7: DYNAMIXEL XL330-M288-T [7]



The CAD models then developed to what is shown in Figure 11. These models allow for four degrees of freedom: shoulder pitch and yaw, elbow flexion, and wrist rotation. Motors housed in the torso and shoulder joint themselves carry out the shoulder's motion. The shoulder is connected to the elbow which features a single motor and is connected to the wrist by the forearm also containing a single motor. The components are fastened together to constrain motion to only allow the desired revolute joints driven by the motors. The motors are supplied with power through wires inside the arm. No sharp edges or pinch points are featured for safety considerations, which meets our design requirements, as does our rounded appearance

To reduce weight, and in turn stress on the motors, the arm shells were made hollow with shells of a thickness of .08 inches (except where thicker portions are needed for functional purposes). When considering the material (PLA, 20-40% infill.), this design principle keeps the arm as lightweight as possible while remaining sturdy enough to not deform from any kind of regular use. It is also consistent with the existing MakiMate design.

To maintain the desired "cute" aesthetic of MakiMate, space for wires was incorporated into the design with holes guiding the wires in paths that would not interfere or hinder the arm while it is in motion. As shown in Figures 8 & 9, the mainly hollow arms and guide holes allow for the wires to be fully encapsulated. Nonexposed wires also reduce risk of users or other outside factors breaking a wire and causing failure. As each of the servo motors are part of the Dynamixel family, the servos can all be directly connected to each other with a daisy chain. There will be a DC-DC 12V to 5V step-down converter in the forearm to provide the proper voltage to the XL330 in the wrist.

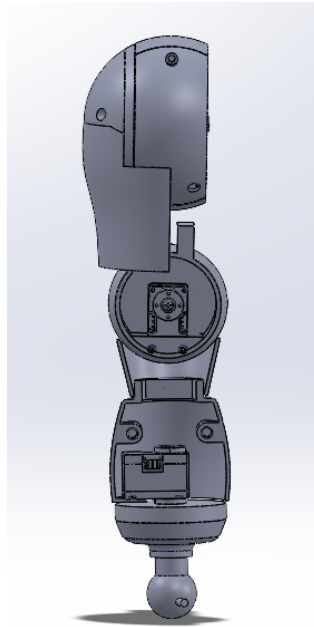


Figure 8: Arm Interior From Side

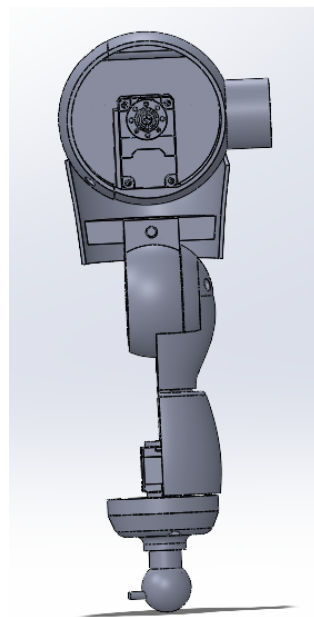


Figure 9: Arm Interior From Front

The pieces were fastened with a combination of M3 x 0.5 brass heat-set inserts (Figure

10), M3 and M2 screws of varying lengths. The heat-set inserts serve the purpose of allowing for disassembly and reassembly while offering better strength and durability. The motors are fastened to their respective mounts with the adequate bolts, with proper dimensioned holes being incorporated into the components of the arm to ensure secure assembly after printing.



Figure 10: M3 x 0.5 Brass Heat-Set Inserts [8]

The components of the arm were designed with 3D printing and ease of assembly in mind. The customer can assemble the components and fasten them together with no difficulty or interferences such as obstructions preventing the use of screwdrivers during assembly. There is also clearance in areas where components touch and experience friction. This was a large factor in the design of the joints, as the nature of 3D printing may produce slight overhang on pieces that can limit their ability to be functional. The pieces of the joints rubbing against each other while moving would produce great friction and prevent smooth movement.

The components of the arm design were constrained together to create an assembly in SolidWorks in accordance with the fastening. The result was an arm design that allowed for only the planned DOF. A motion study simulation was run in SolidWorks that confirmed functionality

of the joints, however the ranges of motion for the joints fell short of the design specifications. This was due to pieces and wiring colliding and interfering with the joint before it reached its goal. The solution was minor revisions in the design of the pieces, smoothing out areas that would allow the joints extra room to achieve a greater range of motion. This proved successful as the updated kinematic simulation displayed ranges of motion that meet or exceed each of the requirements, displaying ranges of 180° , 90° , 80° , and 180° . The resulting design is shown with the full mechanical assembly in Figure 11.

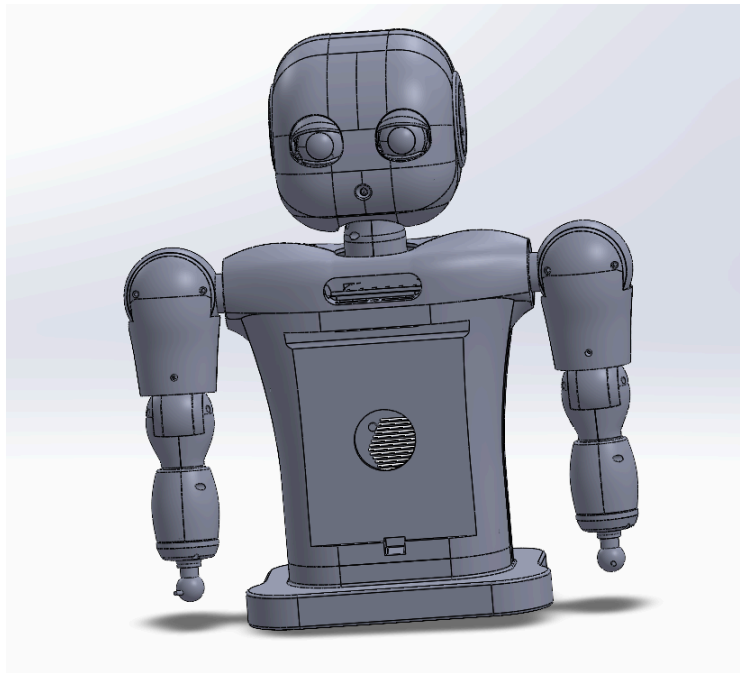


Figure 11: Full Mechanical Design

The characteristics of the arm design allowed for the calculation of the required torque at each of the joints. These calculations were performed with variations of Equations 1 and 2 tailored for the dimensions and weight of the arm components when in the most stressful position (horizontal). The calculations revealed the required design torque at the wrist to be

0.02Nm, which led to the decision of the XL330 to be the wrist's servo motor. The elbow and shoulder yaw required design torques were determined to be 0.27Nm and 0.82Nm, leading to the decision of employing XL430 and XM430 (Figure 12) motors at the elbow and both shoulder joints respectively. The decision process for the determination of the XM430 is shown in Table 3, influencing the decision to select the XL430 and XL330 as they supply the required torque while remaining in the Dynamixel family (allowing for daisy chains). The servo motors possess comfortable general-use bands (20% of stall torque) of 0.82Nm, 0.3Nm, and 0.104Nm for the XM430, XL430, and XL330 respectively, confirming the design specification is met and the motors can move at an adequate speed without overheating or failing.



Figure 12: DYNAMIXEL XM430-W350-T

Table 3: Servo Decision Pugh Matrix				
Criteria	Weight	XL430 (Baseline)	XM430-W350	XM540-W270
Torque	5	0	+4	+5
Size	4	0	0	-4
Weight	3	0	-1	-4

Table 3: Servo Decision Pugh Matrix				
Cost	3	0	-3	-5
Thermal	4	0	+3	+5
Speed	2	0	-1	-3
Weighted Score		0	18	-4

The biggest challenges regarding the arm design lie in the need for proper integration with the other subsystem. This is especially true with the servo motors in the joints, as they require power and communication from the electrical and software subsystems to perform the gestures.

Table 4: Mechanical BOM for MakiMate			
Subsystem	Part	Quantity & UOM	Price
Mechanical Engineering	M3 Threaded Inserts	240	\$7.79
Mechanical Engineering	M5 Threaded Brass inserts	14	\$8.99
Mechanical Engineering	M2 Bolts Assortment Kit	Various	\$7.99
Mechanical Engineering	M2.5 Bolts Assortment Kit	Various	\$7.99
Mechanical Engineering	M3 Bolts Assortment Kit	Various	\$21.99
Mechanical Engineering	M5 (12mm) Bolts	14	\$8.87
Mechanical Engineering	Ball Bearings	1	\$9.49
Mechanical	Metal Dowel Pins	1	\$7.25

Engineering			
Mechanical Engineering	PLA Filament	~3.2Kg	\$80
Mechanical Engineering	TPU Filament	~250g	~\$20

5. Electrical/Electronic Subsystem

The electrical subsystem for MakiMate provides power distribution and connectivity to all computers, sensors, and actuation parts. Unlike the computer/software and mechanical subsystems, the electrical subsystem did not undergo any redesigns this semester. The existing electrical system developed in the previous semester has been evaluated against the updated design requirements required by the other subsystems, and has been determined to be sufficient without modification.

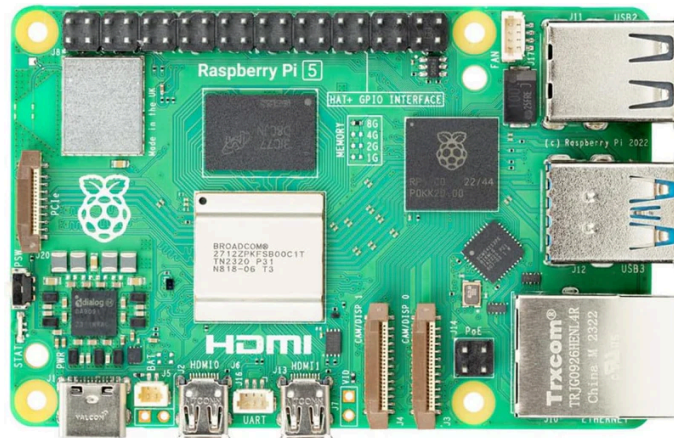


Figure 13: Raspberry Pi 5

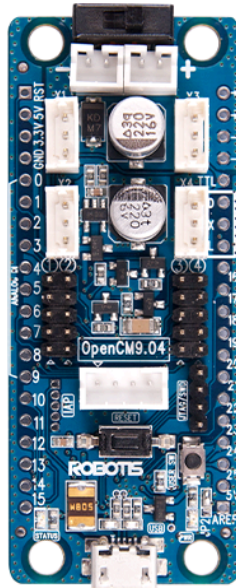


Figure 14: OpenCM9.04

The subsystem employs two controllers: the Raspberry Pi 5 and the OpenCM9.04 microcontroller. The Raspberry Pi 5 executes ROS2-based software responsible for facial tracking and recognition, audio processing, system flow logic, and communication with the LLM. It interfaces with the ReSpeaker USB Microphone array and the Raspberry Pi Camera Module 3 directly, allowing MakiMate to hear and visually track users.

The OpenCM9.04 manages all six Dynamixel XL430-W250-T servo motors responsible for moving the head, eyes, and eyelids. Communication between the Raspberry Pi and the OpenCM9.04 occurs through a USB connection, allowing software to control the motors.

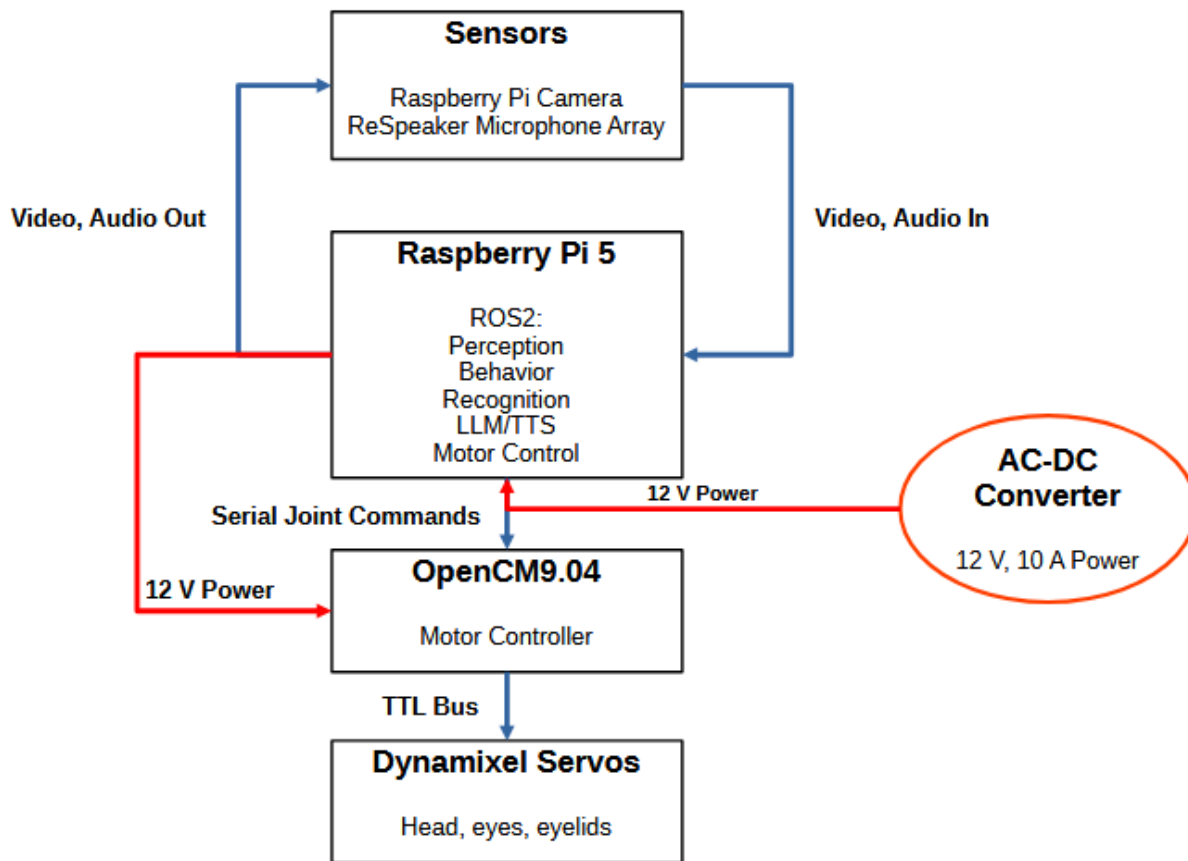


Figure 15: Electrical block diagram illustrating power distribution and communication between the Raspberry Pi 5, OpenCM9.04, sensors, and actuators.

Power is distributed through an AC-to-DC 12 V, 10 A power supply. The OpenCM9.04 and attached servos operate on the 12 V rail directly, ensuring adequate torque. The Raspberry Pi and all 5 V components operate through regulated inputs, minimizing stress. A dedicated 5 V audio amplifier powers the internal speaker, while a USB-to-AUX DAC is used to generate an audio signal from the Raspberry Pi. Reliability is supported by the regulated low-voltage design, reducing the likelihood of overheating.



Wiring within MakiMate uses 22-gauge insulated copper conductors, balancing durability, flexibility, and electrical impedance. Cable routing was designed to separate high-current motor lines from the Raspberry Pi and its components, minimizing electromagnetic interference that could degrade performance. The camera and microphone are positioned to maximize performance while considering aesthetics, cable protection, and ease of transportation.

The subsystem, which is retaining a proven electrical design, supports the updated specifications by providing stable power, with significant headroom given the Pi's maximum capabilities and no additional motors, ensuring safe and reliable communication between controllers.

While the current electrical system is sufficient to support the improvements for MakiMate, several alternatives were evaluated with Pugh Matrices to determine whether or not upgrades to the current subsystem were warranted. However, the existing architecture scored highest overall, and, therefore, remained unchanged.

Table 5: Control Architecture Pugh Matrix					
Criteria	Weight	Baseline	Raspberry Pi 5	Arduino	Jetson Nano
Processing Ability	4	0	4	-4	4
Motor Control	2	0	0	2	0
Software Flexibility	3	0	3	-3	3
Cost	2	0	0	0	-2
Total			7	-5	5

Table 6: Power Architecture Pugh Matrix					
Criteria	Weight	Baseline	Single 12 V Supply	Dual Isolated Supplies	Battery-Based System
Safety	4	0	0	4	-4
Noise susceptibility	2	0	2	2	-2
Thermal Stress	3	0	0	-3	-3
Cost	3	0	0	-3	-3
Maintenance	3		0	0	-3
Troubleshooting	2		2	-2	-2
Total			2	0	-12

Table 7: Vision Sensor Pugh Matrix					
Criteria	Weight	Baseline	Raspberry Pi Camera Module 3	USB Camera	Wireless Camera
Ease of Integration	3	0	3	3	-3
Reliability	3	0	3	3	0
Speed/Latency	4	0	4	0	-4
Cost	2	0	0	0	0
Total			10	6	-7

Major tasks for the electrical subsystem included power budget analysis under worst-case operational conditions and electrical testing during full system integration. Minor tasks included inspection of connectors and cables, verification of grounding and voltage regulation, and thermal monitoring of control boards and motors.

Overall, the electrical subsystem links the computer/software and mechanical subsystems, enabling software to control MakiMate's mechanical motion through stable power



delivery and communication hardware. By utilizing popular and easily accessible off-the-shelf components, the electrical design supports long-term usage, expandability, and ease of implementation.

Table 8: EBOM for MakiMate		
Subsystem	Part	Quantity & UOM
Electrical Engineering	DYNAMIXEL XL430-W250-T	12 EA
Electrical Engineering	DC Power Jack Female Panel Mounting Connector Socket	1 EA
Electrical Engineering	Round Rocker Latching ON/Off	1 EA
Electrical Engineering	22 AWG Solid Core Wire	Various Lengths
Electrical Engineering	DC-DC USB Buck Step Down Module	1 EA
Electrical Engineering	4 pin header	1 EA
Electrical Engineering	2 Pin Connector Plug Wires	3 EA
Electrical Engineering	Aux Cable with Screw Terminals	1 EA
Electrical Engineering	90° USB Cable Extension	1 EA
Electrical Engineering	USB to Aux Out w/ DAC	1 EA
Electrical Engineering	Speaker Stereo	1 EA
Electrical Engineering	5V Audio Amplifier	1 EA
Electrical	Power Supply 12V 10A	1 EA

Engineering		
Electrical Engineering	CS30 RGB-D ToF Depth Sensor	1 EA

6. Computer/Software Subsystem

6.1 Software Architecture Overview

The software subsystem of MakiMate is built on the Robot Operating System 2 (ROS2) framework running on the Raspberry Pi 5. ROS2 was selected for its publish/subscribe messaging architecture, which allows standalone nodes to communicate with one another. This makes it straightforward to develop, test, and replace. The full node graph launched by the presentation mode is shown in Figure 16.

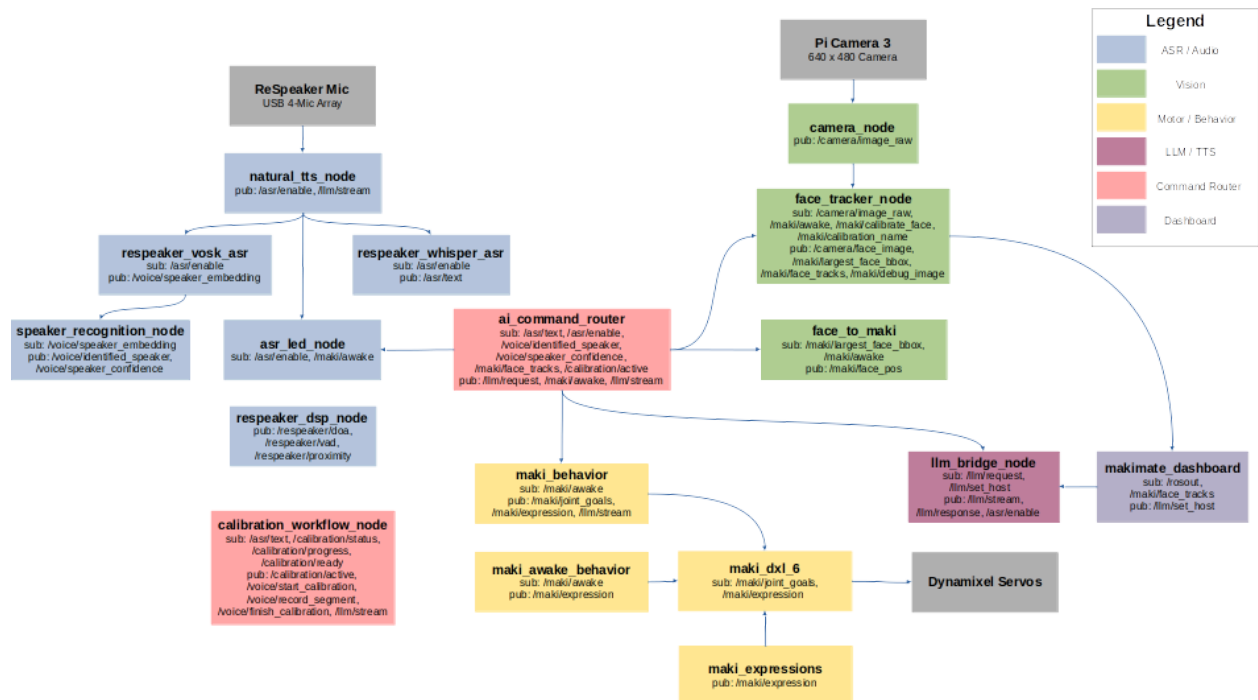




Figure 16: Presentation mode ROS2 node map / flowchart.

The system is organized into five different layers:

1. Sensing. The ReSpeaker microphone array captures audio, and the Raspberry Pi Camera Module 3 captures video. Processing nodes use this raw data directly.
2. Recognition. The speaker recognition node is able to identify the active speaker by comparing live speaker embeddings against stored profiles. Additionally, the face tracker node performs face detection, tracking, and recognition using MediaPipe.
3. Decision making. The ai command router node subscribes to the transcribed user speech, speaker identity, face tracks, and decides how to respond.
4. Response. The llm bridge node forwards requests to the remote LLM server and streams the response back. The natural tts node converts the streamed text to audio and plays it through the speaker. The maki behavior and dxi 6 nodes send behavioral commands and joint goals to the motor controller.
5. Monitoring and control. The makimate dashboard node runs a FastAPI web server on port 8080, providing the user with a configurable, live interface for parameter tuning, starting and stopping the robot, volume control, and LLM/ASR server configuration.

6.2 Large Language Model

The conversational aspect of MakiMate centers around a pipeline which takes user audio input, feeds the question to the Large Language Model (LLM), and then outputs the LLM



response via audio back to the user. This process uses Automatic Speech Recognition (ASR) software Faster Whisper by OpenAI. This software transcribes the user's spoken input from the onboard microphone into a text file which is then sent to the LLM API Ollama. Ollama acts as the middle man between machine computing and the actual thinking model Qwen2.5:3b. The Qwen model will query through its training data and the several databases integrated in the MakiMate system to generate the most appropriate response to the user's inquiry. This text response is then fed into the Text-to-Speech (TTS) software Edge-TTS. Edge-TTS generates an audio file of the LLM's text response. This audio file will then be played on MakiMate's onboard speakers. By looping this process, it is possible to replicate aspects of a natural conversation between a human user and MakiMate. The overarching conversational pipeline is illustrated by Figure 17 below, where the flow of information is visible from the user's input, to the logic, and back to the user as output.

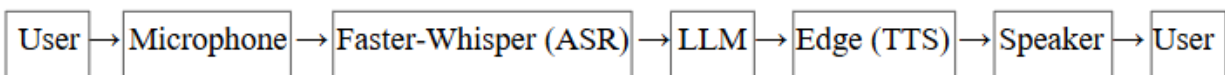


Figure 17. High Level Conversational Pipeline

The following figure details the more complicated process of obtaining the LLM's response to the user. The Python file *chat_llm.py* acts at the main logic center in the entire pipeline. User input via the ASR software is received in this file, from here it will query through the several databases that are integrated into the system for relevant information. That information is then sent to the API. The API is essentially the chat bot interface which allows a user's question to be sent to the thinking model Qwen2.5. Qwen2.5 will take the information from the user and databases provided to it by Ollama and generate a response. The process of

obtaining this response is visualized in Figure 18 below and it will be fed into the TTS software Edge.

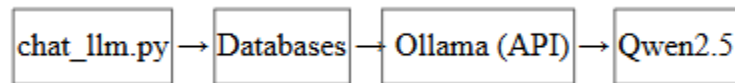


Figure 18. In Depth View of LLM Response Pipeline

There is another relevant python file titled *client.py*. This file is the Python scripting file that is responsible for routing the data from ASR to the LLM pipeline seen in Figure 19 as well as routing the LLM response to the TTS software so it can be played on MakiMate's speakers.

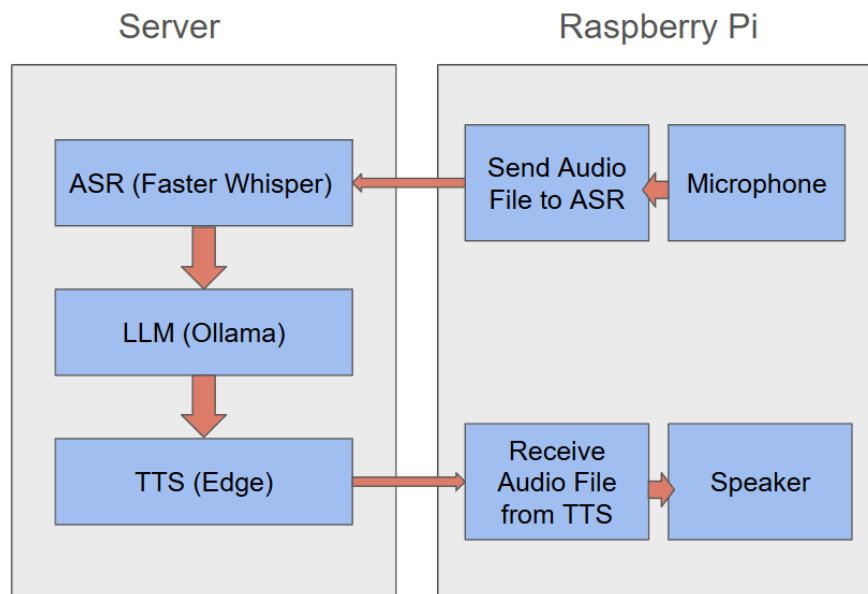


Figure 19. Pipeline Visual

To allow for seamless conversation between the MakiMate and the user, the ASR, LLM, and TTS needed to be blended seamlessly with one another. This was achieved with a python file that integrated all three aspects of the MakiMate conversation pipeline together. The main way it does this is by taking the transcription provided by the ASR and sending it to the LLM server as



a query that the LLM then answers. The ASR is also biased in this file, making lesser known words such as names be registered more accurately by the program. The ASR used in this file is faster-whisper, a smaller, more time-efficient version of the whisper software developed by OpenAI. The decision to opt for a locally hosted, entirely open source conversation pipeline arose after weighing the pros and cons of the other options. This choice would provide the team with the greatest control over what Maki says to users, the latency, and reduced costs to nearly none. The comparisons are quantified with Table 9 below.

Table 9: LLM Pugh Matrix					
Criteria	Weight	Baseline	Cloud LLM	Open Source Local LLM	API
Cost	3	0	3	3	-3
Speed/Latency	4	0	-4	4	4
Reliability	2	0	2	2	-2
Ease of Implementation	1	0	1	1	1
Total			2	10	0

The conversational aspect of the MakiMate needed data on what the robot can talk to a fellow student or guest about. The chat aspect of the Makimate has models that we mentioned above. The way that we were able to provide data to the conversational aspect was to use databases in the python code. The benefit is that it is an organized collection of data that a Python application interacts with to store, retrieve, and manage information. This allows the code to be broken up into smaller folders that the main Chat_LLM python code can access. This allows the code to become a more manageable size to interpret. The group decided that there are



2 likely questions a new student or guest would ask. These questions would be where is a professor's office or where is a lab for a class they are taking. There needed to be 2 databases needed. The research showed that the fastest database to process the requests would be sqlite3. This was due to the fact that it eliminates the network latency. The first database that was built was to locate the professor's office. The thought was to allow the student to say 4 different options to get an answer. The option to address the professor as professor or Doctor. Also to say the professor's complete name or just the last name. ("Doctor Brian Dean", "Doctor Dean", "Professor Dean"). The Makimate will respond with the building and the room number of the professor's office. "446", "Engineering center".

The lab database was built to where the student can ask for the lab's location by 3 options. These were the class number, complete name of class or of a simple shortened nickname for the lab "Robotic Systems and Controls", "robotics lab", "4722". Then MakiMate will give the user that lab room number and building. "346", "Engineering center".

6.3 Hearing – Automatic Speech Recognition

The hearing subsystem is responsible for capturing audio, filtering noise at the hardware level, segmenting speech from continuous audio, and transcribing it to text using Whisper. This subsystem is composed of three nodes that work together: respeaker_dsp_node, respeaker_whisper_asr, and respeaker_vosk_asr.



6.3.1 ReSpeaker DSP Node

The ReSpeaker USB microphone array contains a digital signal processor. Instead of relying on software filtering, the node takes advantage of the onboard DSP, enabling multiple methods of filtering before audio reaches the Pi:

- Stationary and non-stationary noise suppression (STATNOISEONOFF_SR, NONSTATNOISEONOFF_SR): attenuates continuous ambient sounds such as crowd noise or a fan.
- Echo and non-linear echo cancellation (ECHOONOFF, NLATTENONOFF): prevents the robot's own speaker output from being re-captured by the microphone, which would otherwise cause the ASR to transcribe MakiMate's own speech.
- Automatic Gain Control (AGCONOFF): normalizes input level across varying speaker distances.
- High-pass filter at 125 Hz (HPFONOFF = 2): removes low-frequency sounds below the frequency of the human voice.
- Voice Activity Detection threshold (GAMMAVAD_SR): sets the sensitivity of the hardware VAD.

All of these filtering parameters can be adjusted live on the dashboard.

6.3.2 Faster-Whisper ASR Node

The `respeaker_whisper_asr` node converts user speech into text. Previously, the microphone was polled on a two-second interval, which caused truncated words at the



boundaries of utterances, along with transcription failures. This was replaced with a Voice Activity Detection (VAD) approach using webrtcvad.

Audio is captured in 20ms frames. A VadChunker state machine processes each frame by combining speech frames once voice activity is detected, then flushing the buffer to an inference queue after a period of silence (which can be configured via the dashboard). This creates natural utterance boundaries, aligning with how a person pauses between sentences.

An RMS gate parameter provides an additional layer of filtering, discarding audio samples whose root-mean-square fall below the threshold. This prevents distant, quiet audio that the VAD may pass, but is too faint for reliable transcription.

Transcription supports two different modes. The first being local inference (on the Pi) using faster-whisper, with the second being offloading to a more powerful remote server. The former allows ease of use, while the latter lets users decrease latency and use more powerful transcription models. The model size and server URL (if applicable) are configurable via the dashboard.

A post-transcription filter runs on every result before being published to the LLM, preventing hallucinations and false positives from streaming to the LLM. This blocks out noise utterances, such as “huh”, “um” “uh”, etc, and known Whisper hallucinations, such as repetitive phrases that are produced under silent conditions.



6.4 Vision – Face Detection and Recognition

The vision subsystem detects faces in each camera frame, tracks them across frames with identities, scores them for tracking priority, and determines which individual is actively speaking.

6.4.1 Camera

The Raspberry Pi Camera Module 3 is driven by the camera_ros node, which publishes frames to /camera/image_raw at 640x480 resolution. This resolution was found to balance detection accuracy while not overloading the Raspberry Pi 5.

6.4.2 Face Tracker Node

The new face tracker node replaced the old Haar Cascode detector with a new pipeline. Each frame is passed to the MediaPipe FaceLandmarker Model, which detects up to three faces simultaneously and returns 468 facial landmarks per face. Bounding boxes are then derived from the minimum and maximum facial landmarks. Detection is limited to ~10 fps to prevent the Pi from being overloaded. The previous approach detected faces at ~30 fps but provided no landmark data and produced false positives. A tracker then assigns IDs to faces. A 2 second period holds the last bounding box rather than immediately signaling that there is no face, preventing the no-face detection from enabling on missed frames.

Each tracked face receives a score to determine which individual is speaking to makimate:

$$Score = W_c \cdot C + W_s \cdot S + W_l \cdot L + W_m \cdot M$$



Where C is centerness, S is size, L is lifetime, and M is lip motion. All weights are tunable via the dashboard.

Face recognition runs in the background every 3 seconds. Embeddings are compared against a persistent database, and a match is accepted if the Euclidean face distance is below a threshold determined by the dashboards configuration.

A face calibration workflow was implemented to allow a new identity to be registered. When the workflow is initiated, the face tracker stores the embeddings extracted from the camera, allowing the robot to recognize an individual by their face alone.

6.5 Speaker Recognition and Identity Fusion

Knowing who is speaking allows MakiMate to address users by their name during a conversation, which is an essential feature for a social robot, as this makes interactions feel more personal. This is achieved through a fusion of voice and facial recognition.

6.5.1 Voice Speaker Recognition

The `respeaker_asr_node` runs the Vosk speech recognition engine in embedding only mode. This creates an embedding vector for the users' utterance using the Vosk speaker model.

For each incoming embedding, the `speaker_recognition_node` checks for similarities between the new vector provided via the real-time embeddings and the stored speaker profiles. The speaker whose profile provides the highest similarity score above the configured threshold is identified as the active speaker. Otherwise, the speaker is unknown.

Speaker profiles are captured during a calibration session. During calibration, MakiMate asks the user questions in the form of a conversation, and the user replies, allowing MakiMate to capture a wide variety of embeddings while keeping the user engaged.

6.5.2 Identity Fusion

Relying in either voice or facial recognition alone may prove to be unreliable in certain situations. The ai command router node fuses both similarity levels to provide a more confident identity before each interaction with the LLM.

If the fused confidence score meets or exceeds the threshold, the identified name is provided to the LLM and is used in MakiMate's response.

6.6 AI Command Router

The ai command router is the logic node of the entire software subsystem. It subscribes to transcribed speech, voice identity, face tracking, the calibration workflow, and decides how to respond to a given utterance. The node's behavior is defined by a finite state machine with two states.

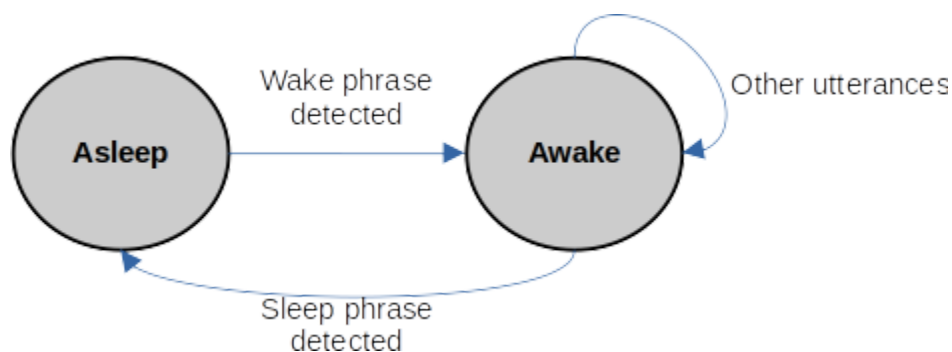


Figure 20. Command Router Visual



In the 'Asleep' state, all incoming utterances, besides the wake phrase, are discarded. When the wake phrase is detected, the robot transitions to the 'Awake' state, and speaks the wake greeting.

In the 'Awake' state, all incoming utterances are processed normally. The node applies the following checks in order:

1. If "calibration" is spoken, the utterance is dropped and the calibration workflow node handles the rest.
2. If the sleep phrase is spoken, the farewell response is spoken, a reset command is sent to the LLM, and the node enters a pending sleep stage, entering sleep afterwards.
3. If the reply keyword is set, utterances not containing that keyword are discarded. This lets MakiMate operate in an environment where it only responds when explicitly addressed.
4. All other utterances are forwarded to and handled by the LLM. Before forwarding, identity recognition is performed and, if a match is found, the identity is provided to the LLM.

6.7 Servo Driver Smoothing

The `maki_dxl_6` node controls the dynamixel servo motors in MakiMate's head via the microcontroller. The previous group's drivers sent position commands directly to the servos at the rate they were received from the behavior nodes, resulting in jerky and erratic movement.

The updated driver node applies exponential moving average smoothing to all of the joint targets:



$$position_smoothed[t] = \alpha \cdot position_target + (1 - \alpha) \cdot position_smoothed[t-1]$$

Where α is the smoothing alpha. This produces a gradual acceleration and deceleration for the motors. This smoothing alpha parameter is adjustable via the dashboard, allowing the user to change the smoothness and speed of the motors.

6.8 Dashboard

The dashboard for MakiMate provides a browser-hosted interface, with it being developed to eliminate lengthy development procedures. Initially, there was a need to access the Raspberry Pi via SSH on a desktop, which required determining the Pi's ip address, which is only accessible through plugging the Pi into a monitor and running commands. Now, the dashboard runs when the Raspberry Pi is plugged in, and can be accessed by any device on the same network using a static url.

The dashboard node runs a FastAPI web application alongside the ROS2 spin loop. Users connect via websocket and receive a snapshot of the current state of MakiMate when they connect. Any further updates are also applied to their dashboard.

The dashboard provides live ROS2 parameter tuning, start/stop for the robot, LLM server URL and model adjustment, ASR server URL and model speaker, volume and microphone gain, git pull and colcon build, and dashboard restart. Additionally, all log messages are forwarded to the dashboard in real time.



6.9 Demo Setup Script

To deploy MakiMate for a demonstration, three separate machines must be configured and connected for minimum latency: the Raspberry Pi running the ROS2 nodes, a GPU server running the Ollama LLM and Faster-Whisper ASR engine, and a desktop acting as a bridge between the connections. Doing this procedure manually required multiple SSH sessions, correctly ordered startup commands, and manually noted ip addresses, which is not user-friendly.

The `setup_demo.sh` script automates the entire process, requiring that only one command must be run in order to use MakiMate.

6.10 Bill of Materials

All software used for MakiMate is free and open-source.

7. Discussion

a. Technical Discussion

The final MakiMate design functions as a completely functional perception, decision, and response system in which all three subsystems work together to support natural interaction. Compared with the original design specifications and proposal, significant progress was made in active-speaker tracking and social interaction. The system improved user tracking through upgraded facial recognition, voice identification, and the addition of software-based noise filtering. The robot's social interaction capabilities were improved through a more responsive and customizable conversational pipeline. These improvements allowed Maki to better maintain attention on the correct user while also strengthening social behaviors such as gaze following and



turn-taking. Additionally, the robotic arm and torso deliverable was advanced through design, analysis, and engineering justification. Elements such as CAD development, torque calculations, tablet integration, and kinematic analysis were all considered and met or exceeded design specifications. In summary, great advancements from the original Maki robot were made, which improved overall system functionality.

b. Professional and Societal Context

i. Engineering Standards

Engineering standards are needed to ensure that the engineering designs are created to a set of formats. These cover multiple areas that an engineer would utilize to create a design. An engineering standard is a set of technical definitions, rules, guidelines, and procedures recognized as authoritative in a specific area of interest. These standards were created to ensure compatibility and uniformity across a variety of industries and organizations. The creation of robots that interact with humans have a set of particular standards to follow. In the context of robotics, standards provide guidelines, which are voluntarily, that promote safety, reliability, and interoperability, while codes are mandatory, legally enforced rules. An example of a relevant code is the National Electric Code (NEC), which governs safe electrical design and installation. On the other hand, patents provide exclusive rights to specific inventions, and reviewing existing patents can help avoid plagiarism.

Since robotics is a broad field, the scope of this search was narrowed down to service robots, a classification that aligns closely with MakiMate's role as a social robot. There are 3 standards and codes that the team found for this classification:



ISO 13482 covers physical safety standards/requirements for robots in human proximity that would apply to MakiMate as it interacts and gestures around people. This standard covers robots that are operated around untrained and vulnerable users as well as children, and carry risks from moving parts (arms, head) that can cause collisions, entanglement, and pinch points [1].

ISO 12100.2 this standard covers the systematic risk assessment and mitigation standards (not robot-specific) and general principles for design. ISO 12100 covers procedures for robots that identify hazards. Estimate and evaluate risks during the life cycle. Eliminate or reduce hazards to the operator or customer [2].

UL 3300 is a standard that states special care is needed to be taken into consideration for robots that are operating around individuals who have disabilities. Covers specific features that a designer may place on robots. This is emphasized on 5 key areas of robot design. Robot design topics are multifrequency outputs for individuals with hearing impairments. Flash rates for people who have epilepsy. Multiple visual indicators for individuals with reduced vision. Using cues or specific colors for individuals with color blindness. Test road conditions that would mimic an individual with a wheelchair or a cane [3].

ii. Safety

In this project, safety mindfulness is crucial, as MakiMate is designed for close interaction with non-technical users in uncontrolled environments. Possible deployments around children, visitors, and individuals with disabilities increase the need for a safe system. Deliverable 1, speaker identification and gaze/head following, directly impacts user comfort and trust. In a social interaction, attention issues can result in feelings of confusion or perceived rudeness. When a robot repeatedly looks away from the active speaker, it may be interpreted as



ignoring them, which can escalate frustration and reduce trust. This becomes particularly important when dealing with vulnerable individuals. As a result, the attention system was developed to reliably focus on the speaker, regardless of a noisy or crowded background. Therefore, mitigating potential risk to the user's physiological well-being.

Deliverable 2 focuses on redesigning the robot's torso and developing new robotic arms, which introduces physical safety concerns related to close human interaction. Since users could be within the reach of the robot, the arm system was designed to reduce impact forces, avoid sharp edges, and move with controlled motion during human interaction. This was addressed by incorporating less powerful servo motors and designing the body with a smooth, rounded geometry. Many non-technical users may not have a complete understanding of the power that robotic arms are capable of. In anticipation of these users, the new mechanical design heavily emphasizes safe usage.

Additionally, changing the torso to add the new arms and an 11-inch display changed the mass distribution of the robot, making stability an important design concern. Over time, repeated loading at the arm mounts could also cause structural problems, which would affect how reliable the system is. If the robot were to tip over, it poses the threat of physically injuring people standing nearby. For this reason, mechanical analysis was used to ensure the new design is capable of safely supporting the robot.

Deliverable 3, modular social behaviors, is also directly tied to the user's welfare and psychological well-being. Maki's behavior layer influences how comfortable and safe the interaction feels for the user. A robot can be physically safe but be mentally harmful if it were to have abrupt behavior, such as interrupting the user. Given the fact that Maki can be deployed with disabled individuals, it is crucial to maintain trust through positive interactions. The



modular framework used supports turn-taking, active listening cues, and faster response timing, helping Maki behave in a more inviting way. The implementation of these modular behaviors will reduce the chance that the robot will be perceived in a negative manner during demonstrations.

iii. Ethical Considerations

The main ethical considerations in the development of the MakiMate robot stem from its use of camera vision, audio capture, and conversational technology while closely interacting with human users. Since the robot relies heavily on a camera and a microphone, user privacy is a concern. As a result, the system was developed with an understanding that user audio, facial data, and recorded interactions should be handled with care, ensuring it is not leaked. Another ethical concern is transparency; users need to understand that MakiMate is a system intended to assist and interact with, but cannot supplement human judgment. While great, artificial intelligence models can provide incorrect information or have lapses in judgment, which could have a potentially harmful impact. Overall, ethical responsibility in this project meant designing MakiMate in a way that is respectful, safe, and appropriate for the real-world environments it will be deployed in.

iv. Economic Factors

Economic factors remained a high priority throughout the design process because MakiMate is intended to be reproducible within a typical low-scale budget and maintainable without the need for highly specialized or costly equipment. For this reason, the project aimed to improve the robot's abilities without unnecessary costs. Deliverable 1 managed to improve user tracking only through software-based methods, introducing no additional financial burden. In



contrast, Deliverable 2 required additional physical components. Cost effectiveness was maintained by prioritizing 3D printed structural components and researching optimal servo motor selection. Since 3D printing is relatively accessible and well-suited for prototyping, it provides a practical and economical means of manufacturing.

Similar to deliverable 1, deliverable 3 provides economic value by relying purely on a software-based solution. To add on to this further, previous groups have used OpenAI's ChatGPT API as their AI backbone, which varies in cost depending on the model of ChatGPT being used. The price of a text input varies between \$1-\$5 across OpenAI's various models, while audio input easily costs between \$10-\$32 per input on models such as gpt-realtime [1]. Given the need to rent computing power from OpenAI, this approach is quite expensive per use. The new proposed solution would use an open-source pipeline which consists of the Text-to-Speech (TTS) software Edge, the Speech-to-Text (STT) software Faster Whisper, and Dialogue Manager Nodes to house the decision logic. All of this software is available for free and the heavy computing power necessary for the LLM is housed on a server owned by the Oakland University School of Engineering. These factors combined dramatically reduce the price of ongoing operation. Therefore the cost of operation for the new conversational pipeline which was developed this semester is virtually zero, with the only exception being the price of electricity needed for the server.

v. Reliability

Reliability needs to be taken into account for the creation of our robot. Businesses are judged on how well their items are built and how long they last. When designing the MakiMate robot, we need to balance cost and materials. This is a topic in the magnitude of engineering



design throughout the world. We could make the robot out of high-tech materials, although the downside of that is the cost of the item. How do engineers test their design? They analyze their design in 3 main ways. These are the Failure Mode and Effects Analysis (FMEA), Rigorous Stress and Accelerated Life Testing, and Redundancy and Factors of Safety.

As mentioned previously, the lack of reliance on an external company such as OpenAI makes ensuring solid performance a more manageable task. The open-source nature of the conversational system ensures that it will perform as expected once the code is finalized. The only issues that could arise would be due to unexpected situations such as a power failure on the university grid which would shut down the server. However, there are measures in place meant to prevent loss of data in the event of a power outage.

vi. Aesthetics

MakiMate is no longer a rough lab prototype; it has now progressed beyond a proof of concept design. As a result of the improvements made this semester, the robot now has a clear humanoid form and interactive features. Its expressive eye and head movement, integrated speech/vision electronics, and torso/arm redesign give it a more polished feel when interacting with a user. Because the MakiMate platform is open source, it does not fit under the same category as typical “for sale” products. Rather, the marketing approach for this robot could be one that targets students and researchers by highlighting its effectiveness, ease of use, and low cost. One could sell already assembled and functional robots to said students/researchers to further increase the ease of use for this product. However, to do so, future teams must address a few shortcomings. First, the torso/arm design currently remains theoretical, not actually having been mechanically assembled. The complete addition of this would expand Maki’s



communication capabilities, therefore increasing customer interest. Additionally, the robot is not prepared for large-scale manufacturing. As it stands, the current design does not consider the tolerances or material selections required for mass production.

vii. Potential Customers

MakiMate is designed to serve a wide range of users by combining an approachable, child-friendly appearance with interactive robotic capabilities. Its soft, friendly design makes it especially well-suited for hospitals and medical environments, where a conversational robot can help reduce anxiety and provide comfort—particularly for children and individuals with socialization challenges. Beyond healthcare, MakiMate can function as an interactive greeter in environments such as schools, hotels, and restaurants. In these settings, it can be customized to provide location-specific information, assist visitors, and enhance overall user experience. MakiMate also has strong potential as an educational tool. It can be deployed in schools and robotics programs as a hands-on teaching aid, giving students practical exposure to robotics, programming, and AI systems in an engaging and accessible way.

What additional areas can MakiMate be incorporated into? MakiMate has strong potential for future deployment in elder care and therapeutic support environments, where consistent, patient, and human-friendly interaction is essential. In assisted living facilities and home care settings, MakiMate can serve as a companion robot that helps reduce loneliness while also providing practical support such as medication reminders, daily schedule prompts, and basic wellness monitoring. Its approachable, child-friendly design can make it less intimidating than traditional technology, encouraging more natural interaction from elderly users who may be unfamiliar with advanced digital systems. In addition to elder care, MakiMate is well-suited for



therapy and behavioral support applications, particularly for individuals with autism spectrum disorder or other socialization challenges. The robot can provide structured, predictable interactions that help users practice communication skills, emotional recognition, and social behaviors in a low-pressure environment. It can also be used in speech therapy or guided behavioral exercises, reinforcing learning through repetition and engagement. By combining emotional accessibility with programmable functionality, MakiMate offers a scalable solution that supports both caregivers and therapists, while improving the quality of care and interaction for users in these sensitive and high-impact environments.

viii. Societal and Global Impact

MakiMate has the ability to benefit society by supporting applications in education, research, assistive interactions, and other settings where interactive robots can engage with individuals. The improvements made in this robot, including active-speaker tracking, modular social behaviors, and a more expressive physical design, make Maki more effective and inviting. The use of this robot could provide advancements in the educational development of children or individuals with learning disabilities. Additionally, it could provide valuable insight into human-robot interaction, which is especially relevant with all of the artificial intelligence advancements being made in today's day and age. Potential drawbacks include privacy concerns, uncomfortable interactions, and a risk that this technology may not be available to those in underprivileged communities. On a global scale, the use of social robotics in public environments raises the issue of cultural adaptability. Some cultures may be more accepting of



this technology than others. Overall, the technology developed through MakiMate could have a positive impact on society if advanced and used responsibly.

c. Information Literacy

Throughout the development of MakiMate, the team had to learn many topics independently that were not taught in any coursework. This includes areas such as ROS 2, speaker identification, large language models, databases, and new mechanical design software. To solve technical issues, the team relied on gathering information from official MakiMate documentation, GitHub repositories, internet resources, and recurring feedback from mentors. These sources of information were chosen based on relevance, reliability, and whether they could realistically support the implementation of the project's new features. All information was used responsibly by properly acknowledging prior work, documenting major revisions, and validating decisions rather than relying on assumptions. This project reinforced the fact that engineering not only depends on one's current knowledge but also on one's ability to continuously learn and apply information to solve unfamiliar problems.

8. Conclusions and Recommendations

In summary, the final MakiMate design produced a more capable social robotics platform by improving the robot's user tracking and responding more naturally in conversation. Additionally, future senior design teams have been provided a solid foundation for the application of a new mechanical design of arms and tablet integration into the torso. This project



successfully addressed the challenges of extending an existing open source social robotics platform into a more practical system by combining mechanical, electrical, and software-based solutions. One of the most difficult aspects of the project was system integration. Coordinating real time vision, audio, conversational elements, and motor control proved to be no easy task. This project also showed that some goals, including the physical redesign, require more time than anticipated. For future teams, next steps would include completing the physical implementation of the robotic arms and tablet-integrated torso, further improving Maki's conversation elements such as response latency, and possibly continuing to improve the robot's physical appearance. Another concern for future groups will be the current power supply. With the additional motors from the arms and the depth camera, we believe more power will be required to operate the system properly.

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10. Appendices

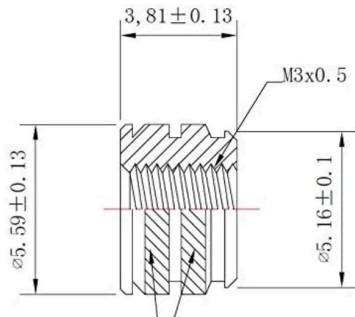


Figure 1: Brass Insert Dimensions